

Algebra II with Trigonometry

Chapter 6.3

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4531393_alg-2trig-h-chp-6-3-note.pdf](#)

Here is a complete study guide and summary based on the provided document.

1. Summary of the Document

This document is a guided lesson on graphing trigonometric functions, specifically sine and cosine. It introduces the concept of periodic functions and establishes the base graphs for $y = \sin t$ and $y = \cos t$ using a unit circle and a 5-point graphing method (Middle, High, Low). The core of the lesson focuses on graphing transformations of these functions by identifying changes to the amplitude, period, horizontal shift (phase shift), and vertical shift. It concludes with a framework for applying these concepts to graphing calculators and real-world word problems (like the vibrations of a tuning fork).

2. Core Definitions, Theorems, and Formulas

Definitions:

- **Periodic Function:** A function f is periodic if there is a positive number p such that $f(t + p) = f(t)$ for every t .
- **Period:** The least such positive number p (if it exists) is the period of f . It represents the length of one complete cycle of the graph.
- **Amplitude:** The number $|a|$, which represents the largest value these functions attain from their midline.

Core Formulas & Transformations: For the general shifted sine and cosine curves (where $k > 0$):

$$y = c + a \sin k(x - b)$$

$$y = c + a \cos k(x - b)$$

- **Amplitude:** $|a|$
- **Period:** $\frac{2\pi}{k}$
- **Vertical Shift:** c
- **Horizontal Shift:** b
- **Graphing Interval:** To graph one complete period, use the interval $[b, b + \frac{2\pi}{k}]$.
- **Horizontal Stretching/Shrinking:** The value of k shrinks the graph horizontally if $k > 1$, and stretches the graph horizontally if $k < 1$.

The 5-Point Sinusoidal Patterns: (Where M = Middle, H = High, L = Low)

- $+\cos t = H M L M H$
- $-\cos t = L M H M L$
- $+\sin t = M H M L M$
- $-\sin t = M L M H M$

3. Worked Study Guide: Step-by-Step

Step 1: Understanding the Base Graphs Before shifting anything, you must know what basic sine and cosine waves look like. Both have a standard period of 2π .

- **Sine** starts at the midline (0), goes to its highest point (1), back to the midline (0), down to its lowest point (-1), and back to the midline (0).
Pattern: M H M L M.
- **Cosine** starts at its highest point (1), goes to the midline (0), down to its lowest point (-1), back to the midline (0), and up to its highest point (1).
Pattern: H M L M H.

Step 2: Identifying the Transformations When you look at an equation like $y = 3 + 2 \sin 4(x - \frac{\pi}{2})$, identify your a , b , c , and k values based on the formula $y = c + a \sin k(x - b)$:

- $c = 3$ (**Vertical Shift**): The whole graph moves up by 3. Your "Middle" (M) line is now $y = 3$ instead of the x-axis.
- $a = 2$ (**Amplitude**): The graph stretches vertically. From your midline of 3, your "High" (H) points will be $3 + 2 = 5$, and your "Low" (L) points will be $3 - 2 = 1$.
- $k = 4$ (**Used for Period**): The period is $\frac{2\pi}{k}$. Here, $\frac{2\pi}{4} = \frac{\pi}{2}$. The graph completes one full wave in $\frac{\pi}{2}$ units along the x-axis. Because $k > 1$, this graph is horizontally *shrunk*.
- $b = \frac{\pi}{2}$ (**Horizontal Shift**): The graph shifts to the right. Instead of starting at $x = 0$, your first point starts at $x = \frac{\pi}{2}$.

Step 3: Sketching One Complete Period To sketch the graph accurately, you need a starting x-value and an ending x-value. The document provides the interval formula: $[b, b + \frac{2\pi}{k}]$. Using our example above:

- Start at $b = \frac{\pi}{2}$.
- End at $b + \frac{2\pi}{k} = \frac{\pi}{2} + \frac{\pi}{2} = \pi$.
- Your 5 sinusoidal points (M H M L M for positive sine) will be evenly spaced between $x = \frac{\pi}{2}$ and $x = \pi$.

4. Practice Problems

Problem 1: Identify the amplitude, period, vertical shift, and horizontal shift for the function: $y = -4 + 5 \cos 2(x - \pi)$

Problem 2: State the appropriate mathematical interval $[b, b + \frac{2\pi}{k}]$ on which to graph one complete period of the function: $y = \sin 3(x + \frac{\pi}{6})$

Problem 3: A tuning fork's vibrations are modeled by the function $v(t) = 0.8 \sin(660\pi t)$. Based on the formulas in the text, what is the period of this vibration?

Problem 4: If you are sketching the graph of $y = -3 \sin(x)$, what 5-point sequence (using M, H, and L) should you use, and why?

Problem 5: Does a k value of $\frac{1}{2}$ stretch or shrink the graph horizontally compared to the base graph?

Answers to Practice Problems

- **Answer 1:** Amplitude = 5 (from $|5|$); Period = π (from $\frac{2\pi}{2}$); Vertical shift = -4 ; Horizontal shift = π .
- **Answer 2:** First, identify $b = -\frac{\pi}{6}$ and $k = 3$. The period is $\frac{2\pi}{3}$. The interval is $[-\frac{\pi}{6}, -\frac{\pi}{6} + \frac{2\pi}{3}]$. By getting common denominators ($-\frac{\pi}{6} + \frac{4\pi}{6}$), the final interval is $[-\frac{\pi}{6}, \frac{\pi}{2}]$.
- **Answer 3:** The k value is 660π . The period formula is $\frac{2\pi}{k}$. Therefore, the period is $\frac{2\pi}{660\pi} = \frac{1}{330}$ seconds.
- **Answer 4:** You would use **M L M H M**. You use this because it is a *negative* sine function, which reflects the standard positive sine wave upside down.
- **Answer 5:** Because $k = \frac{1}{2}$, which is less than 1 ($k < 1$), it **stretches** the graph horizontally.

5. Assumptions and Ambiguities Resolved

- **Missing Definition for Frequency:** In Example 6(b), the document asks to find the "frequency" of a vibration but does not explicitly provide the formula for frequency in the text. I assumed this was background knowledge expected by the teacher ($\text{Frequency} = \frac{1}{\text{Period}}$), but strictly adhered to your prompt to base practice problems *only* on the document. Therefore, I omitted frequency from the practice problems and focused heavily on the period formula which *is* provided.
- **Typo in Example 6:** The original text says "...where is the displacement of the times in millimeters at time seconds." This is a typo in the provided worksheet. It almost certainly means "...where v is the displacement of the *tines* in millimeters at time t seconds." I resolved this by treating the function $v(t)$ standardly as displacement over time.

- **Blank Examples:** Examples 2 through 6 in the document contain blank spaces intended for a student to fill out during class. Because I cannot provide graphs using external plotting software (and to keep everything strictly contained to the provided formulas), my summary and practice problems focus on the *algebraic and methodological* rules the document gives to solve those types of problems (like finding a , b , c , and k).